26.05.2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 50	Examination: ESE Examination	Duration: 2 Hrs.
Programme code: 01 Programme: B. Tech COMP	Class: TY	Semester: V (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 16U01C502	Name of the Course: Computer Networks	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	PWD - SECTION B	Max. Marks
Q 4	Solve any **four** (Short notes / Short question type)	20
i)	Write a short note on DHCP.	5
ii)	Write a short note on IGMP.	5
iii)	Write a short note on TCP Three-way Handshake.	5
iv)	Compare HTTP and FTP	5
v)	IP is connectionless and best-effort. Justify the statement.	5
vi)	Write a short note on Telnet..	5

Que. No.		Question [Section B]	Max. Marks
Q 5	Solve any **Two**		20
A	i)	Explain how ARP works when the source and destination hosts are in different IP networks.	5+5
	ii)	Include the ARP Request and Response packet formats used in this process. Support your answer with a diagram.	
B	i)	Explain the **Distance Vector Routing Algorithm** used in network layer routing.	5+5
	ii)	Discuss the **Count-to-Infinity problem** associated with it. Illustrate your explanation with a suitable example and describe at least one technique used to overcome this problem.	
C	Explain the **TCP state transition diagram** with respect to **connection establishment** and **termination**.		5+5

Que. No.	Question	Max. Marks
Q 6	Answer the following:	20
i)	What is the purpose of the PSH flag in the TCP header? A. Typically used to indicate end of message B. Typically used to indicate beginning of message C. Typically used to push the message D. Typically used to indicate stop the message	2M

ii)	If a computer is unable to access the Internet, but can communicate with local network devices, what should be checked first? A. Router configuration B. Cable Connections C. Local Firewall Settings D. DNS Settings	2M
iii)	What is the maximum size of data that the application layer can pass on to the TCP layer below? A. Any size B. 216 bytes – size of TCP header C. 216 bytes D. 1500 bytes	2M
iv)	In the slow start phase of the TCP congestion control algorithm, the size of the congestion window A. does not increase. B. increases linearly. C. increase quadratically. D. increase exponentially.	2M
v)	The protocol data unit (PDU) for the application layer in the Internet stack is A. segment. B. datagram. C. Message D. frame.	